

CS5275 – The Algorithm Designer's Toolkit
(S2 AY2025/26)

Lecture 13:

Boolean functions – Arrow's impossibility theorem

Condorcet paradox

- **Majority rule:**
 - If most people prefer a over b , then a wins.
- But what happens when there are more than 2 candidates?





Condorcet paradox

- **Majority rule:**
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- But what happens when there are more than 2 candidates?

Setup: 3 voters rank 3 candidates: a, b, c

- Voter 1: $a > b > c$
- Voter 2: $b > c > a$
- Voter 3: $c > a > b$

Majority rule

Condorcet cycle:

- a vs. b : a wins.
- b vs. c : b wins.
- c vs. a : c wins.



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Majority rule

Condorcet cycle:

- a vs. b : a wins.
- b vs. c : b wins.
- c vs. a : c wins.

- **Question:** Can we avoid Condorcet cycles by using other rules?

Impossibility



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Can we design a voting system that always converts individual preferences into a fair group decision?

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- Arrow proved a surprising result.
 - If there are more than 2 candidates, no voting rule can satisfy all of the following:

Pareto efficiency:

- If everyone prefers a over b , the group should prefer a over b .



Unanimous:

$$f(1, \dots, 1) = 1 \text{ and } f(-1, \dots, -1) = -1$$

Impossibility



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Independence of irrelevant alternatives:

- The ranking between a and b should depend only on how people rank a vs. b .

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Take-away messages:

- Designing a perfectly fair voting system is impossible.
- Trade-offs between fairness principles are unavoidable.

Impossibility

Arrow later received the **Nobel Memorial Prize in Economic Sciences**, partly for this work.

<https://www.nobelprize.org/prizes/economic-sciences/1972/arrow/facts/>



This insight helped create the field of **social choice theory**.

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 - If there are more than 2 candidates, no voting rule can satisfy all of the following:

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Take-away messages:

- Designing a perfectly fair voting system is impossible.
- Trade-offs between fairness principles are unavoidable.

Setting

	Voters' preferences					Outcome
	#1	#2	#3	...		
$a(1)$ vs. $b(-1)$	1	1	-1	...	$= x$	$f(x)$
$b(1)$ vs. $c(-1)$	1	-1	1	...	$= y$	$g(y)$
$c(1)$ vs. $a(-1)$	-1	-1	1	...	$= z$	$h(z)$

Independence of irrelevant alternatives:

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For simplicity, in the lecture we assume $f = g = h$.

We leave the extension to allow different functions as an **exercise**.

Independence of irrelevant alternatives:

- The ranking between a and b should depend only on how people rank a vs. b .

Setting

$$\text{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

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$b \text{ (1) vs. } c \text{ (-1)}$	1	-1	1	...	$= y$	$f(y)$
$c \text{ (1) vs. } a \text{ (-1)}$	-1	-1	1	...	$= z$	$f(z)$

There are $3! = 6$ possible rankings of 3 candidates, corresponding to 6 possible values of (x_i, y_i, z_i) :

$(1, 1, -1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, 1, 1), \quad (1, -1, 1), \quad (-1, -1, 1)$

These are precisely the triples satisfying the **not-all-equal** predicate NAE_3 .

Setting

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Condorcet paradox:

- $f = \text{Maj}_n$ does not work.

A candidate is a **Condorcet winner** if it wins all of the pairwise elections in which it participates.

$$\exists \text{ Condorcet winner} \iff \text{NAE}_3(f(x), f(y), f(z)) = 1$$

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Arrow's theorem: Suppose $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is a unanimous voting rule.

- If there is always a Condorcet winner, then f must be a dictatorship.

Proof idea:

- Analyze $\Pr[\exists \text{ Condorcet winner}] = \mathbb{E}[\text{NAE}_3(f(\mathbf{x}), f(\mathbf{y}), f(\mathbf{z}))]$

Analysis

$$\text{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

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Condorcet paradox:

- $f = \text{Maj}_n$ does not work.

Each voter chooses its ranking uniformly at random, over the 6 possible rankings of 3 candidates:

$$(x_i, y_i, z_i) \sim \{(1, 1, -1), (1, -1, -1), (-1, 1, -1), (-1, 1, 1), (1, -1, 1), (-1, -1, 1)\}$$

Analysis

$$\text{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

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Condorcet paradox:

- $f = \text{Maj}_n$ does not work.

Each voter chooses its ranking uniformly at random, over the 6 possible rankings of 3 candidates:

$$(x_i, y_i, z_i) \sim \{(\textcolor{brown}{1}, \textcolor{brown}{1}, -1), (\textcolor{violet}{1}, -\textcolor{violet}{1}, -1), (-\textcolor{violet}{1}, \textcolor{violet}{1}, -1), (-\textcolor{violet}{1}, \textcolor{violet}{1}, 1), (\textcolor{violet}{1}, -\textcolor{violet}{1}, 1), (-\textcolor{brown}{1}, -\textcolor{brown}{1}, 1)\}$$

$(x, y), (y, z), (z, x)$ are $(-1/3)$ -correlated.

(x, y) is a ρ -correlated pair of random strings:

- $x \sim \{-1, 1\}^n$
- $y \sim \{-1, 1\}^n$
- $\mathbb{E}[x_i y_i] = \rho$

Analysis

$$\text{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

$\Pr[\exists \text{ Condorcet winner}]$

$$= \mathbb{E}[\text{NAE}_3(f(\mathbf{x}), f(\mathbf{y}), f(\mathbf{z}))]$$

$$= \frac{3}{4} - \frac{1}{4} (\mathbb{E}[f(\mathbf{x})f(\mathbf{y})] + \mathbb{E}[f(\mathbf{y})f(\mathbf{z})] + \mathbb{E}[f(\mathbf{z})f(\mathbf{x})]) \quad \leftarrow \text{NAE}_3(s_1, s_2, s_3) = \frac{3}{4} - \frac{1}{4} (s_1 s_2 + s_1 s_3 + s_2 s_3)$$

$$= \frac{3}{4} \left(1 - \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \text{ } (-1/3)\text{-correlated}}[f(\mathbf{x})f(\mathbf{y})] \right)$$

Each voter chooses its ranking uniformly at random, over the 6 possible rankings of 3 candidates:

$$(\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i) \sim \{(1, 1, -1), (1, -1, -1), (-1, 1, -1), (-1, 1, 1), (1, -1, 1), (-1, -1, 1)\}$$

$(\mathbf{x}, \mathbf{y}), (\mathbf{y}, \mathbf{z}), (\mathbf{z}, \mathbf{x})$ are $(-1/3)$ -correlated.

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Analysis

$$\begin{aligned} & \Pr[\exists \text{ Condorcet winner}] \\ &= \mathbb{E}[\text{NAE}_3(f(\mathbf{x}), f(\mathbf{y}), f(\mathbf{z}))] \\ &= \frac{3}{4} - \frac{1}{4} (\mathbb{E}[f(\mathbf{x})f(\mathbf{y})] + \mathbb{E}[f(\mathbf{y})f(\mathbf{z})] + \mathbb{E}[f(\mathbf{z})f(\mathbf{x})]) \\ &= \frac{3}{4} \left(1 - \mathbb{E}_{(\mathbf{x}, \mathbf{y})}^{(-1/3)\text{-correlated}}[f(\mathbf{x})f(\mathbf{y})] \right) \end{aligned}$$

Lemma: (when $-1 < \rho < 0$)

$$\mathbb{E}_{(\mathbf{x}, \mathbf{y})}^{\rho\text{-correlated}}[f(\mathbf{x})f(\mathbf{y})] = \rho$$

f is a dictator or its negation

$$\text{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

$$\Pr[\exists \text{ Condorcet winner}] = 1$$

f is a **dictator** or its negation.

Impossible because f is **unanimous**.

Pareto efficiency

$(\mathbf{x}, \mathbf{y})$ is a ρ -correlated pair of random strings:

- $\mathbf{x} \sim \{-1, 1\}^n$
- $\mathbf{y} \sim \{-1, 1\}^n$
- $\mathbb{E}[x_i y_i] = \rho$

Noise stability

$$\Pr[\mathbf{y}_i = \mathbf{x}_i] = \frac{1}{2}(1 + \rho) \quad \Pr[\mathbf{y}_i = -\mathbf{x}_i] = \frac{1}{2}(1 - \rho) \quad \longleftrightarrow$$

Noise stability of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ at ρ is
$$\text{Stab}_\rho(f) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \text{ } \rho\text{-correlated}}[f(\mathbf{x})f(\mathbf{y})].$$

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Noise stability

$$\text{Stab}_\rho(\chi_S) = \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[\chi_S(x)\chi_S(y)] = \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[\prod_{i \in S} x_i y_i] = \prod_{i \in S} \mathbb{E}[x_i y_i] = \rho^{|S|}$$

Noise stability of $f : \{-1,1\}^n \rightarrow \mathbb{R}$ at ρ is
 $\text{Stab}_\rho(f) = \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[f(x)f(y)]$.

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$$\text{Stab}_\rho(\chi_S) = \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[\chi_S(x)\chi_S(y)] = \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[\prod_{i \in S} x_i y_i] = \prod_{i \in S} \mathbb{E}[x_i y_i] = \rho^{|S|}$$

$$\begin{aligned} \text{Stab}_\rho(f) &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[f(x)f(y)] \\ &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}} \left[\sum_{S, T \subseteq [n]} \hat{f}(S)\hat{f}(T)\chi_S(x)\chi_T(y) \right] \\ &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}} \left[\sum_{S \subseteq [n]} \hat{f}(S)^2 \chi_S(x)\chi_S(y) \right] \\ &= \sum_{S \subseteq [n]} \hat{f}(S)^2 \rho^{|S|} \end{aligned}$$

Noise stability of $f : \{-1,1\}^n \rightarrow \mathbb{R}$ at ρ is
 $\text{Stab}_\rho(f) = \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[f(x)f(y)]$.

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Noise stability

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$$W^k[f] = \sum_{S \subseteq [n] : |S|=k} \hat{f}(S)^2$$

Noise stability

$$\begin{aligned}
 \text{Stab}_\rho(f) &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[f(\mathbf{x})f(\mathbf{y})] \\
 &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}} \left[\sum_{S, T \subseteq [n]} \hat{f}(S)\hat{f}(T)\chi_S(\mathbf{x})\chi_T(\mathbf{y}) \right] \\
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 &= \sum_{S \subseteq [n]} \hat{f}(S)^2 \rho^{|S|} = \sum_k \rho^k \cdot W^k[f]
 \end{aligned}$$

$$W^k[f] = \sum_{S \subseteq [n] : |S|=k} \hat{f}(S)^2$$

$$\begin{array}{c}
 \sum_k W^k[f] = 1 \\
 \downarrow \\
 \text{Equality holds iff } W^1[f] = 1 \\
 \downarrow \\
 \sum_k \rho^k \cdot W^k[f] \geq \rho \cdot \sum_k W^k[f] = \rho \\
 \uparrow \\
 \rho^k \text{ is minimized when } k = 1.
 \end{array}$$

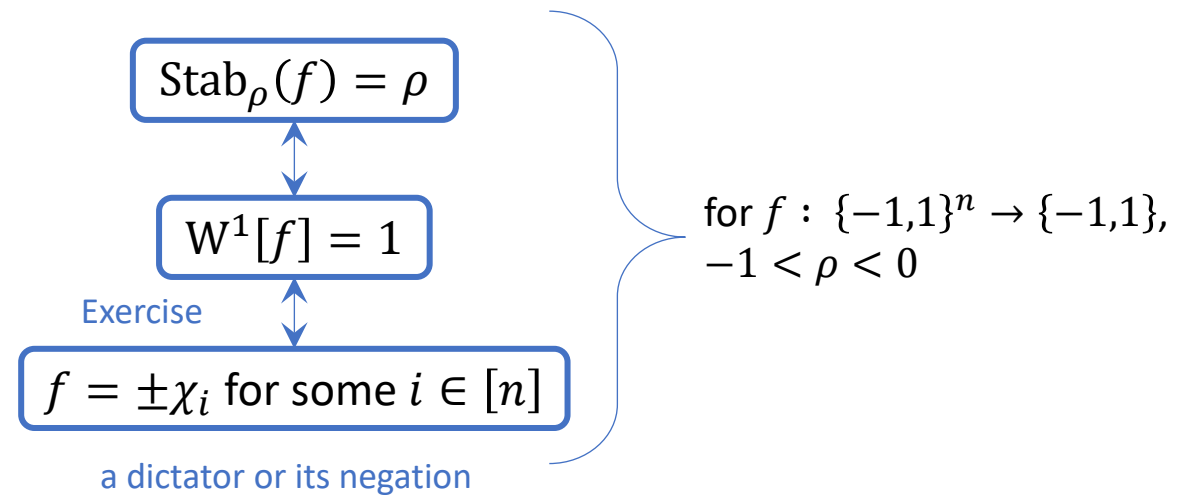
$$\left. \begin{array}{c} \text{Stab}_\rho(f) = \rho \\ \updownarrow \\ W^1[f] = 1 \end{array} \right\} \begin{array}{l} \text{for } f : \{-1,1\}^n \rightarrow \{-1,1\}, \\ -1 < \rho < 0 \end{array}$$

Noise stability

$$\begin{aligned}
 \text{Stab}_\rho(f) &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}}[f(x)f(y)] \\
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 &= \mathbb{E}_{(x,y) \text{ } \rho\text{-correlated}} \left[\sum_{S \subseteq [n]} \hat{f}(S)^2 \chi_S(x)\chi_S(y) \right] \\
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$$W^k[f] = \sum_{S \subseteq [n] : |S|=k} \hat{f}(S)^2$$

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 \sum_k W^k[f] = 1 \\
 \downarrow \text{Equality holds iff } W^1[f] = 1 \\
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 \end{array}$$



Beyond the impossibility result

Getting further insight via Fourier analysis

$$\lim_{n \rightarrow \infty} \text{Stab}_\rho(\text{Maj}_n) = 1 - \frac{2}{\pi} \arccos(\rho)$$



Using $f = \text{Maj}_n$:

- $\lim_{n \rightarrow \infty} \Pr[\exists \text{ Condorcet winner}] = \frac{3}{2\pi} \arccos\left(-\frac{1}{3}\right) \approx 91.2\%$

Guilbaud's formula

Beyond the impossibility result

Getting further insight via Fourier analysis

$$\lim_{n \rightarrow \infty} \text{Stab}_\rho(\text{Maj}_n) = 1 - \frac{2}{\pi} \arccos(\rho)$$



The bound attained by the majority function is already the best possible.

Using $f = \text{Maj}_n$:

- $\lim_{n \rightarrow \infty} \Pr[\exists \text{ Condorcet winner}] = \frac{3}{2\pi} \arccos\left(-\frac{1}{3}\right) \approx 91.2\%$

Using any $f : \{-1,1\}^n \rightarrow \{-1,1\}$ with all $\hat{f}(\{i\})$ equal:

- $\lim_{n \rightarrow \infty} \Pr[\exists \text{ Condorcet winner}] \leq \frac{7}{9} + \frac{4}{9\pi} \approx 91.9\%$
- $\lim_{n \rightarrow \infty} \Pr[\exists \text{ Condorcet winner}] \leq \frac{3}{2\pi} \arccos\left(-\frac{1}{3}\right) \approx 91.2\%$

Chapter 5.4

Chapter 11.7

Upper bound for any **transitive-symmetric** function.

Robustness of the impossibility result

$\Pr[\exists \text{ Condorcet winner}] \in 1 - \Theta(\epsilon)$

Exercise

$W^1[f] \in 1 - \Theta(\epsilon)$

Friedgut–Kalai–Naor (FKN) theorem

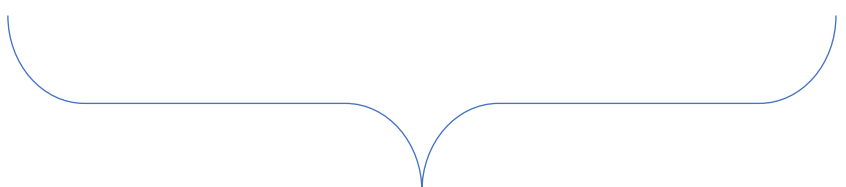
f is $\Theta(\epsilon)$ -close to $\pm\chi_i$ for some $i \in [n]$

Outlook

- **Next:** spectral structure + learning.

How to learn an unknown Boolean function via queries?

What structures enable efficient learning?



Boolean functions with sufficiently simple Fourier spectra can be efficiently learned.

References

- Sections 2.4–2.5 of <https://arxiv.org/abs/2105.10386>